

JOANNE TANG

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EDUCATION

National University of Singapore

Aug 2019 - Present

Bachelor of Engineering (Mechanical Engineering), Honours

- Current Grade Point Average (GPA): 3.75/5.0
- Expected Graduation: Jul 2023

CentraleLaVie

Feb 2021 – Sep 2021

Student Exchange Program in France

Design Project: Optimization of Chassis for 3D printing

- Utilized topology optimization and additive manufacturing principles, resulting in cost reduction
- Applied FEA to ensure chassis can withstand multiple load cases
- Repositioned critical electronics in device to reduce maintenance time by 10%
- Prepared a cost benefit analysis to evaluate feasibility of redesign

WORK EXPERIENCE

Shell

Jun 2020 - Jan 2021

Mechanical Engineering Intern

- Designed a calibration tool that improved measurement accuracy, accompanied with tolerance stack analysis and engineering drawings
- Designed a pipeline that conformed to Health, Safety and Environment (HSE) requirements. Performed cost analysis that facilitated supplier selection, achieved targeted cost savings
- Utilized FEA to analyze structural integrity which led to more than 10% in cost savings. Supported in a patented method of sealing pressure based on performed analysis
- Improved design calculations by incorporating industrial standards which increased accuracy by 5%

Magnitude

May 2019 - Jul 2019

Hardware Engineering Intern

- Designed, conducted, and reported impact test experiments on a chassis, identified failure prone areas leading to an improvement in toughness of chassis
- Remodeled chassis by improving opening mechanism reduced time to access main electronics by 20%
- Designed and 3D printed a fixture, improving manufacturing time by at least 50%

PROJECTS AND COMPETITIONS

Final Year Project: Computational Stress Analysis of Hybrid Composites

Jul 2021 - Present

- Analyzed stress distributions of various parameters using Finite Element Analysis (FEA)
- Proposed and validated a hypothesis using FEA to understand failure mechanism

Design Project: 3D Printed Soft Robotic Hand

Feb 2021 - Jun 2021

- Designed, controlled, and tested robotic hand using SimLab simulation software
- Analyzed and optimized physical parameters of robotic finger using robot kinematics in MATLAB, resulted in a 15% increase in workspace

EY Sustainability Challenge

Dec 2018

- Developed an algorithm to analyze more than 10,000 datasets and records to optimize energy

usage for an energy conservation challenge

- Achieved top 10 standing amongst 105 teams

SKILLS

- Languages: Proficient in English and Chinese (spoken and written), basic spoken Bahasa Indonesia and French
- CAD packages: Solidworks (Certified SolidWorks Associate - Mechanical Design), Creo/Pro-E
- FEA packages: Ansys, Abaqus
- Mathcad, Matlab, Microsoft Project, Basic Programming (C)

EXTRA-CURRICULAR ACTIVITIES

NUS RC4 (Residential College 4), Community Lead

Feb 2018 - Apr 2019

- Led a committee of 3 members to organise 5 community building activities involving more than 200 residents, successfully integrated seniors, exchange students and freshmen.

Anderson Junior College Guitar Ensemble

Feb 2017 - Nov 2017

Vice President

- Spearheaded a Community Involvement Project at St Andrew's Hospital, reaching out to 20 seniors through interaction and playing guitar pieces
- Volunteered to be in an organizational team for performing arts concert, showcasing 3 guitar pieces, achieving a turnout rate of roughly 800 people
- Led 8 committee members to organize a 3D2N camp for more than 50 juniors

INTERESTS

Baking and playing the acoustic guitar